

NBA Database Documentation

PART 1: Table Documentation

Table 1: team

Purpose:

Contains core identification details for each NBA franchise.

Key Columns:

- Primary Key: id – A unique value assigned to every team.
- Foreign Keys: None. This table serves as a foundational reference for other tables.
- Name/Identifier Columns:
 - full_name (e.g., 'Los Angeles Lakers')
 - abbreviation (e.g., 'LAL')
 - nickname (e.g., 'Lakers')
 - city (e.g., 'Los Angeles')
 - state (e.g., 'California')
- Data/Stat Columns:
 - year_founded – Indicates the year the franchise was formed.
- Relationships:
 - team.id is used by common_player_info.team_id to associate players with teams.
 - game.team_id_home and game.team_id_away reference team.id to determine competing teams.
 - team_info_common.team_id links team data to seasonal statistics.

Table 2: common_player_info

Purpose:

Holds in-depth player information including background details, physical measurements, team affiliation, and career data.

Key Columns:

- Primary Key: person_id – Uniquely identifies each player.
- Foreign Keys: team_id – Connects to team.id to show current team membership.
- Name/Identifier Columns:
 - first_name
 - last_name
 - display_first_last

- position (e.g., Guard, Forward)
- school
- country
- Data/Stat Columns:
 - height
 - weight
 - season_exp
 - jersey
 - draft_year
 - draft_round
 - draft_number
- Relationships:
 - common_player_info.team_id joins to team.id.
 - common_player_info.person_id is referenced by draft_history.person_id.

Table 3: inactive_players

Purpose:

Lists players who were inactive for specific games due to injuries or coaching decisions.

Key Columns:

- Primary Key: game_id + player_id – Composite key for inactive entries.
- Foreign Keys:
 - game_id – References game_info.game_id.
 - player_id – References player.id.
 - team_id – References team.id.
- Name/Identifier Columns:
 - first_name
 - last_name
 - team_city
 - team_name
 - team_abbreviation
- Data/Stat Columns:
 - jersey_num
- Relationships:
 - inactive_players.game_id links to game_info.game_id.
 - inactive_players.player_id links to player.id.
 - inactive_players.team_id links to team.id.

Table 4: player

Purpose:

Stores simplified player identification data and active status indicators.

Key Columns:

- Primary Key: id – Distinct identifier for each player.
- Foreign Keys: None. This table acts as a base reference.
- Name/Identifier Columns:
 - full_name
 - first_name
 - last_name
- Data/Stat Columns:
 - is_active – Shows whether the player is currently active in the league.
- Relationships:
 - player.id is referenced by inactive_players.player_id.
 - player.id can be linked to common_player_info.person_id for expanded player details.

Table 5: team_history

Purpose:

Records franchise history including relocations and team name changes over time.

Key Columns:

- Primary Key: team_id + year_founded – Composite key per franchise era.
- Foreign Keys:
 - team_id – References team.id.
- Name/Identifier Columns:
 - city
 - nickname
- Data/Stat Columns:
 - year_founded
 - year_active_till
- Relationships:
 - team_history.team_id links back to team.id.

Table 6: officials

Purpose:

Tracks referees assigned to NBA games and connects officials to specific matchups.

Key Columns:

- Primary Key: game_id + official_id – Composite key ensuring one entry per official per game.
- Foreign Keys:
 - game_id – References game.game_id.
- Name/Identifier Columns:
 - official_id
 - first_name
 - last_name
- Data/Stat Columns:
 - jersey_num – Official's jersey number.
- Relationships:
 - officials.game_id links to game_info.game_id.
 - Multiple officials are recorded per game.

Table 7: team_details

Purpose:

Provides extended franchise details including arena information, management staff, and online presence.

Key Columns:

- Primary Key: team_id – Matches team.id.
- Foreign Keys: team_id – References team.id.
- Name/Identifier Columns:
 - abbreviation
 - nickname
 - city
 - arena
 - owner
 - generalmanager

- headcoach
- dleagueaffiliation
- Data/Stat Columns:
 - yearfounded
 - arenacapacity
- Social Media Columns:
 - facebook
 - instagram
 - twitter
- Relationships:
 - team_details.team_id extends team.id with one-to-one data.
 - Arena data can be joined with game results for venue-based analysis.

Table 8: game_info

Purpose:

Stores scheduling and attendance-related metadata for NBA games.

Key Columns:

- Primary Key: game_id – Unique game identifier.
- Foreign Keys: None directly.
- Name/Identifier Columns:
 - game_id
 - game_date
- Data/Stat Columns:
 - attendance
 - game_time
- Relationships:
 - game_info.game_id is referenced by officials.game_id.
 - game_info.game_id is referenced by inactive_players.game_id.

PART 2: Query Ideas

- How many players in the database are currently active versus inactive or retired?
(uses player table)
- Which official has refereed the highest number of NBA games?
(uses officials table)
- What is the highest recorded attendance for any NBA game?
(uses game_info table)
- Which NBA team plays in the arena with the largest seating capacity?
(uses team_details table)
- Which NBA franchises have relocated the most times throughout their history?
(uses team_history and team tables)
- Which team has had the greatest number of inactive players across all games?
(uses inactive_players and team tables)